

Wk2.2 Ch2 wrap and Ch3 intro

Thursday, September 17, 2026 9:05 AM

Big idea: $Y = f(X) + \epsilon$, $X = (X_1, \dots, X_p)$

2.1.1: RIE: $E[(Y - \hat{Y})^2] = (f(X) - \hat{f}(X))^2 + \text{Var}(\epsilon)$

2.1.2: BVT: $E[(Y_0 - \hat{Y}_0)^2] = (\text{Bias}(\hat{f}(X_0)))^2 + \underbrace{\text{Var}(\hat{f}(X_0))}_{\text{small}} + \text{Var}(\epsilon)$

2.1.3: $\text{Bias}(\hat{f}(X_0)) = E[\hat{f}(X_0)] - f(X_0)$

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Last Ch2 topic: So far, only considered regression ($-\infty < Y < \infty$), but same framework applies to classification ($Y \in \{c_1, c_2, \dots, c_k\}$)

$Y = f(X) + \epsilon$, $Y \in \{0, 1\}$ or (Ch4) 3+ class

Red/irred & Bin/var both are same.

Trade-off test error: MSE can't all missclassif. Yes = 0 No = 1

$\text{PCE}_{\text{train}} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(Y_i \neq \hat{Y}_i)$

$\mathbb{I}(Y_i \neq \hat{Y}_i) = \begin{cases} 1, & Y_i \neq \hat{Y}_i \\ 0, & Y_i = \hat{Y}_i \end{cases}$

2 classifiers: ① Bayes classifier (but best) ② k-nearest neighbors (KNN) Simple! Read

Ch3 Regression

2 parts: ① Simple linear regression (X single variable) ② Multiple linear regression ($X = (X_1, \dots, X_p)$)

Simple linear regression

$Y = f(X) + \epsilon$

Assume $f(X) = \beta_0 + \beta_1 X$, then $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$

Goal: Estimate $\hat{\beta}_0$ & $\hat{\beta}_1$ via minimizing error

error $\rightarrow e_i = Y_i - \hat{Y}_i$ (residuals)

via $\text{RSS} = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 \geq 0$

Gives an optimization problem, 2 ways to solve: ① Calc 1 (diff) ② LA (FROLA)

Calc 1: Minimize $\text{RSS} = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i))^2 \geq 0$

$\frac{\partial \text{RSS}}{\partial \hat{\beta}_0} = 0 \rightarrow \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)) = 0$

$\frac{\partial \text{RSS}}{\partial \hat{\beta}_1} = 0 \rightarrow \sum_{i=1}^n X_i (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 X_i)) = 0$

Solve for $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$

In end best fit line $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$ has

$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$

$\hat{\beta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$

LA System

$n \text{ eqns } \begin{cases} \beta_0 + \beta_1 X_1 = Y_1 \\ \vdots \\ \beta_0 + \beta_1 X_n = Y_n \end{cases} \rightarrow \begin{bmatrix} 1 & X_1 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}$

$A \vec{\beta} = \vec{y}$ (soln)

FROLA

$\vec{\beta} = (A^T A)^{-1} A^T \vec{y}$

Finding $\hat{\beta}_0$ & $\hat{\beta}_1$ is easy (2 formulas), but care about stat inference (certainty)

stat 195 $\uparrow$ CI: $\hat{\beta}_1 \pm 1.96 \text{SE}(\hat{\beta}_1)$ 95% $\rightarrow$ smish

2 types: HT $\hat{\beta}_1 \neq 0$ p-val $\rightarrow$ smish